LAST NAME: _____

# Practice Quiz 09D — MATH 2055 — Fall 2026

Practice for Quiz 09

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

*Revised using GPT-6 Astra in the Codex harness.*

1. (a) Define $f : \{1, 2\} \to \{a, b, c\}$ by $f(1) = a$ and $f(2) = b$. Choose $Y \subseteq \{a, b, c\}$ such that $f(f^{-1}(Y))$ is a proper subset of $Y$, and verify your choice by computing the sets.

   (b) Prove by induction that $3 \mid (4^n - 1)$ for every integer $n \geq 0$. Here $3 \mid m$ means $m = 3k$ for some integer $k$.